7. Non-Primitive SJF Algorithm

Code:

```c
#include <stdio.h>

int main()
{
    int n, i, j, time = 0, temp;
    int bt[20], at[20], wt[20], tat[20], p[20];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        p[i] = i + 1;
        printf("Enter Arrival Time and Burst Time of P%d: ", i + 1);
        scanf("%d%d", &at[i], &bt[i]);
    }

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (at[j] < at[i])
            {
                temp = at[i]; at[i] = at[j]; at[j] = temp;
                temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
                temp = p[i]; p[i] = p[j]; p[j] = temp;
            }
        }
    }

    for (i = 0; i < n; i++)
```

```
    {
        int min = i;

        for (j = i + 1; j < n; j++)
        {
            if (at[j] <= time && bt[j] < bt[min])
                min = j;
        }

        if (at[min] > time)
            time = at[min];

        temp = bt[i]; bt[i] = bt[min]; bt[min] = temp;
        temp = at[i]; at[i] = at[min]; at[min] = temp;
        temp = p[i]; p[i] = p[min]; p[min] = temp;

        wt[i] = time - at[i];
        time += bt[i];
        tat[i] = time - at[i];

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    printf("\nProcess\tAT\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t%d\t%d\n",
               p[i], at[i], bt[i], wt[i], tat[i]);

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

    return 0;
}
```

Output:

```
Enter number of processes: 3
Enter Arrival Time and Burst Time of P1: 4
3
Enter Arrival Time and Burst Time of P2: 2
3
Enter Arrival Time and Burst Time of P3: 2
3

Process AT      BT      WT      TAT
P2      2       3       0       3
P3      2       3       3       6
P1      4       3       4       7

Average Waiting Time = 2.33
Average Turnaround Time = 5.33
```